

Q3. Worked Example Document Classification (based on slide: Test document “predictable no fun”)

Using the smoothed likelihoods and priors from Q2, compute the probability scores for the document “*predictable no fun*” under both the positive and negative classes.

Priors: $P(-) = 3/5$, $P(+) = 2/5$

Tasks:

1. Show each step of the multiplication.

Denominators:

$$\text{Negative} = 14 + |V| = 14 + 20 = 34$$

$$\text{Positive} = 9 + |V| = 9 + 20 = 29$$

Smoothed likelihoods (from the slide):

$$\text{Negative : } P(\text{Predictable} | -) = 1 + 1/34 = 2/34,$$

$$P(\text{no} | -) = 1 + 1/34 = 2/34,$$

$$P(\text{fun} | -) = 0 + 1/34 = 1/34$$

$$\text{Positive: } P(\text{Predictable} | +) = 0 + 1/29 = 1/29$$

$$P(\text{no} | +) = 0 + 1/29 = 1/29$$

$$P(\text{fun} | +) = 1 + 1/29 = 2/29$$

2. Which class should the system assign to this document?

$$\text{Score } (-) = 3/5 * 2/34 * 2/34 * 1/34$$

$$= 6.1 * 10^{(-5)}$$

$$\text{Score } (+) = 2/5 * 1/29 * 1/29 * 2/29$$

$$= 3.3 * 10^{(-5)}$$

Since $\text{Score}(-) > \text{score}(+)$, we can assign the document to negative class.